ATLAS V4 toe and jet angle tests, 21 Oct 2026

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Keep the outer jetfoils straight and set PX4's hover pitch to the aircraft's natural angle (about 17.7°): that gives 91 % of simulated flights passing with no hardware change. The jets must really leave at 67.5° or a little higher; if they leave lower, fix the angle mechanically.

Setup

All tests use V4 as built: 3 batteries, 10–13 kg, your bench thrust curve (extended to 32–37 N at full throttle), jets at 67.5°. Each setup flew the same 64 random aircraft in PX4 simulation: park at −20°, rotate, take off, climb to 5 m, full-stick yaw, pitch and roll. A flight passes if it takes off, stays within 6° of the commanded attitude and recovers. Almost every aircraft took off in every test, so the failures below happen in the air.

Tests 1–2: turning the outer jetfoils (toe)

The outer pair was turned from −30° (thrust inward) to +30° (outward), first with PX4 hovering at 20°, then with the hover pitch moved to each setup's natural angle (trim).

Toe	Pass at 20°	Pass with trim	Yaw, worst case (with trim)
−30°	91 %	88 %	0.68 rad/s²
−10°	80 %	88 %	0.82 rad/s²
0° (as built)	75 %	**91 %**	**1.04 rad/s²**
+10°	47 %	91 %	0.95 rad/s²
+30° *	0 %	91 %	1.43 rad/s²

* At +20° and +30° the outer jet comes within 2–5 cm of the middle jetfoil: likely not buildable.

Without the trim, toe only trades flights for control. With the trim, every toe from 0° up passes 91 %, so toe is not needed; outward toe adds some yaw but is limited by space.

Tests 3–4: what if the air does not leave at 67.5°?

The jet angle is a design value; the real airflow may follow the jetfoil more or less (Coanda). We moved all six jets together, keeping toe 0° and PX4's hover pitch fixed at 17.7°, then re-trimmed two angles.

Real jet angle	63°	64.5°	65.5°	66.5°	**67.5°**	68.5°	69.5°	70.5°
Pass, trim fixed	16 %	28 %	62 %	86 %	**91 %**	83 %	75 %	64 %
Pass, re-trimmed		56 %					91 %	

Every 1° of jet error moves the natural hover angle by about 1°, so with a fixed trim the aircraft leans more and fails. A higher jet is fully fixed by re-trimming (69.5° back to 91 %). A lower jet is not: at 64.5° the jets point more forward, a full roll leaks into yaw and about one roll in three departs, whatever the trim.

What it means

- **The layout is right as built:** toe 0°, jets 67.5°, battery where it is.
- **The hover-pitch trim is what makes it fly:** it takes the pass rate from 75 % to 91 % at no cost.
- **The trim must come from the real aircraft,** not the CAD, because it follows the real jet angle.
- **A low jet is the one real risk:** below about 66°, software cannot fix it.

What to do from now on

1. **Build as is:** outer jetfoils straight, jets 67.5°, 3 batteries; aim for 13 kg or less.
2. **Check the real jet angle on one jetfoil:** smoke or tufts, or a bench force measurement (also thrust to 100 % and fan torque). If it is below about 66°, adjust the rear fins up toward 67.5° before flying.
3. **Tethered hover:** start PX4 at a hover pitch of 17.7°, measure the steady lean, and move the hover pitch to remove it (about −1° per +1° of jet angle).
4. **Short hops, then the full manoeuvres.**
5. **Later, for more yaw:** counter-rotating fan pairs, or +10° outward toe after a CAD clearance check.

Not simulated: thrust above 60 % throttle (extended from your bench points), structural strength, ground effect, wind and transition to cruise.